HUMAN FACTORS, 1982, 24(5), 509–519

Why Alphabetic Keyboards Are Not Easy to Use: Keyboard Layout Doesn't Much Matter

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These studies demonstrate the inferiority of alphabetically organized keyboards as compared with a randomly organized keyboard and the standard Sholes (qwerty) keyboard. Use of the alphabetic keyboard requires considerable mental processing; the novice is faced with a trade-off between mental processing and visual search, and this makes different keyboard layouts equivalent. Comparison of different keyboard layouts by computer simulation of expert typing shows surprisingly little effect of keyboard arrangement for a wide class of keyboards. Performance with some alphabetical layouts is quite slow, but with others, it is within 2% of the speed achieved when using the Sholes keyboard. Performance with the Dvorak keyboard is only improved by about 5% over performance on the Sholes keyboard. The conclusion is that it is not worthwhile to use alphabetic keyboards for novice typists, nor to change to the Dvorak layout for experts. Keyboards can probably be improved, but only through radical redesign of the present physical key configuration.

It seems obvious that the traditional typewriter keyboard—the "qwerty" or "Sholes" keyboard—presents many difficulties for novice typists. The arrangement of the letters on the keyboard seems arbitrary and difficult to learn. The keys were organized by the Sholes brothers in 1873 to minimize the jamming of type bars in their early design of the typewriter. They placed the keys that were typed successively as far apart on the keyboard as possible, so that the type bars would approach each other at a relatively sharp angle, thus minimizing the chance of jamming. Ease of learning or of typing were not considered; another triumph of technology over human factors (see Beeching, 1974). There appears to be no system to the layout, and beginners often ask why the keys cannot be laid out in alphabetical order. A number of typewriter-like devices do arrange the keys in alphabetical order, including children's electronic toys, hand-held language translators, note-taking devices, and a number of commercial machines and stockbroker displays.

Although an alphabetical arrangement might be best for novices, different considerations are relevant for expert typists. Here, one wishes to lay out the keys so as to maximize typing rate. One major redesign of the keyboard is known as the "Dvorak keyboard." This is a keyboard arrangement based upon human factors (time-and-motion study) principles. The arrangement emphasizes an efficient layout of the keys to minimize hand and finger motion (Dvorak, 1943; Dvorak, Merrick, Dealey, and Ford, 1936). Proponents of the Dvorak keyboard

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have frequently demonstrated advantages in learning time and typing speed, but to little avail against the established dominance of the Sholes arrangement. The question to be examined here is: How big a difference does keyboard layout make for novice and expert?

POSSIBLE ADVANTAGES OF ALPHABETIC KEYBOARDS

Alphabetical ordering of the keys makes logical sense, and early keyboards were indeed alphabetical. The qwerty arrangement was invented to minimize jamming of keys, but jamming is not a major concern with modern type bars and typing elements, electric typewriters and computer terminals. An alphabetical arrangement would make sense to inexperienced typists, who today must spend considerable time learning the arbitrary arrangement of the Sholes keyboard. Perhaps the strongest argument against any proposed change is simply that the Sholes keyboard is so well known and used that it would not be practical to change. The argument is persuasive enough to have defeated attempts to implement the Dvorak keyboard. The strongest argument for alphabetically arranged keyboards applies to the performance of the novice typist. Both the Sholes and Dvorak keyboards seem arbitrary and, for the novice, there appears to be no reason for the location of the letters on the keyboard. Even the arrangement of keys is peculiar, with several rows of diagonally structured keys that are not well designed for the structure of the hands, and shift and return keys located in ways that make typing awkward. Many people have experimented with other schemes (Alden, Daniels, and Kanarick, 1972; Litterick, 1981), but major change seems unlikely, except for specialized application. But even if the physical structure of the keyboard is taken as given, why not arrange the keys into a structured format so that the novice can find them?

Is the alphabetical arrangement of keys superior for novices? It seems obvious that the answer must be yes; how could it be otherwise? Surely, knowledge of the alphabet would aid in finding the keys on the keyboard, thus enhancing typing speed and reducing mental workload for the novice. However, studies of alphabetic keyboards fail to support this view; novice typists type as rapidly on the Sholes keyboard as on alphabetically organized keyboards (Hirsch, 1970; Michaels, 1971; also see the review by Alden et al., 1972). These studies seem to contradict common sense. Most people believe that the alphabetic arrangement of keys should be superior—at least for novices. Why then have previous studies not shown the superiority?

One possible answer is that alphabetic keyboards have been compared with the Sholes keyboard. It is difficult to find people who have not had some exposure to the qwerty keyboard arrangement, and so it is possible that the results were biased by past experience. Informal observation of operators of a plastic card-making machine provides support to the notion that knowledge of the Sholes arrangement interferes with the use of alphabetic keyboards, as these operators exhibited considerable difficulty with the alphabetically arranged keys. It would seem that they were not much helped by the alphabetical arrangement, and that for those who knew anything about typing (with the Sholes keyboard), this knowledge was interfering. If knowledge of the Sholes keyboard interferes with the use of alphabetic ones, then in a study of keyboards, the proper control would be to use a randomly organized keyboard, since it (unlike the Sholes keyboard) is *completely* unfamiliar to the novice typist. Accordingly, we set out to retest the

differences among keyboards, using a randomly structured keyboard as our control.

EXPERIMENTAL ANALYSIS OF ALPHABETIC KEYBOARDS

Method

The four keyboards shown in the top half of Figure 1 (a, b, c, and d) were tested with subjects who classified themselves as nontypists. The Sholes keyboard was a standard computer keyboard that was connected to one of the laboratory computers. The other three keyboards were made by taking three older keyboards and moving the key caps to make the desired arrangement. These keyboards were connected to a set of lights (LEDs) that displayed the code of the key being pressed, but they were not otherwise operational. (These keyboards were left over from a previous generation of discrete transistor circuits and were not easily interfaced with the voltage levels now in use in the laboratory.) All keyboards were mounted in a sloping panel of relatively standard dimensions, height, and angle, and the three non-qwerty keyboards were identical except for the arrangement of their key caps.

The interkeystroke intervals for the Sholes keyboard were recorded by the computer. Because the other keyboards could not be connected to our system, we videotaped the hand and eye positions, using a counter that electronically put a number on each video frame. The videotape was analyzed with a stop-action recorder and video disc, allowing for reasonably rough determination of keystroke intervals (to within 1/60 s). We started the

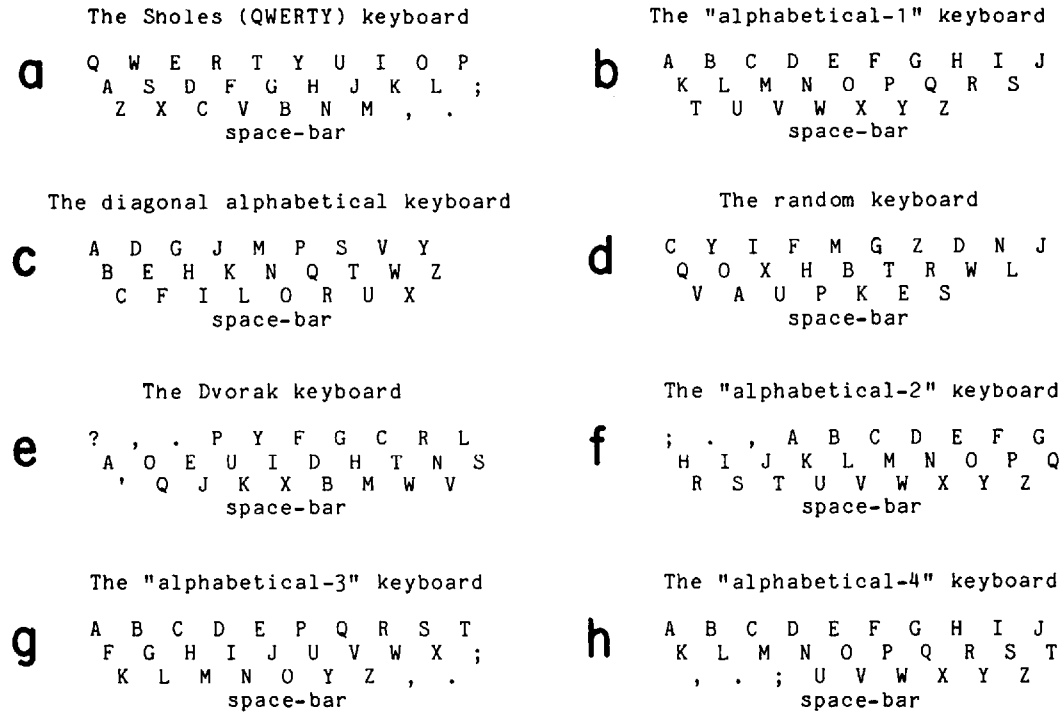

Figure 1. *Six experimental keyboard arrangements (A, B, C, and D for Experiment 1; A, B, C, D, E, F, G, H for the simulation.*

timing when the LEDs went on, since the LED code gave an unambiguous indication of which key was depressed.

Subjects. Twelve subjects were studied, all undergraduate students at the University of California, San Diego, who either received course credit for taking part in the experiment or who were paid. The subjects were screened for typing ability; the advertisement had requested that only nontypists apply. All subjects were given a pretest on the Sholes keyboard and any who typed over 25 words/min (2.1 letters/s) were dismissed. The average typing speed of the 12 subjects on the Sholes keyboard was 16 words/min (1.3 letters/s).

The speed criterion (25 words/min) is the same as that used by Hirsch (1970). The mean typing rate of his "nontypists" was 0.9 letters/s, compared with our value of 1.3.

Procedure. The subjects were first given a 10-min pretest on the Sholes keyboard. They were then tested for 10 min on each of the other three keyboards in counterbalanced order, with a 5-min rest between tests. (With three keyboards and 12 subjects, this yielded two replications of each order.) The same text was used throughout; we opted for consistency of the letter strings in the experimental material at the cost of some learning of the material, counting on the counterbalancing to even out the factors. The text was adapted from a *Reader's Digest* article on diets. It was approximately 12 000 characters long and was presented as double-spaced, typewritten copy, printed in all uppercase characters. The design of the keyboard was explained to each subject prior to each trial. The instructions stressed speed with accuracy. Subjects typed as much of the text as they could within the 10 min allowed them. After all the trials had been completed, we asked the subjects for their impressions and preferences.

Experimental Results and Discussion

The mean numbers of letters typed by the subjects during each 10-min session, excluding punctuation and spaces, are presented in Table 1. The keystrokes were determined by viewing the videotapes and noting the LED indicators activated by the keyboards. Note that subjects did not type very much in each 10-min session; most subjects only managed to complete from 60 to 100 words with the alphabetic and random keyboards. Most errors were substitutions of words similar in sounds or meanings to the target words. Errors also included skipped words. Thus, most typing errors were perceptual or memorial in nature; they were not keyboard typing errors. (Typing rates are usually computed by dividing the number of letters typed per minute by five, the standard conversion between letters and words. Spaces are typed with relative ease, and do not differ across keyboards. Thus, by not counting the typing of spaces, we amplify any possible differences among keyboards. When qwerty typing speeds are computed including the spaces, the mean speed increases from the 13 words/min shown in Table 1 to 16 words/min.)

TABLE 1

Letters Typed in 10 Minutes by Nontypist Subjects (Not Including Spaces and Punctuation)

	KEYBOARDS			
	Sholes	*Alphabetical-1*	*Diagonal*	*Random*
Mean	648	388	391	356
Letters/s	1.08	0.65	0.65	0.59
Words/min	13.0	7.8	7.8	7.1

An analysis of variance performed on the non-qwerty keyboards showed a significant effect of keyboard type, $F(2,22) = 6.24$, $p < 0.01$). A 95% Sheffé confidence interval on the data (with order used as a covariate) showed any difference of 29 keystrokes to be statistically significant. Thus, there is a small (10%) but significant superiority of the two alphabetical keyboards over the random one, but no difference between the two alphabetic types. The times for the qwerty keyboard are included for comparison purposes; they are clearly superior. (The qwerty keyboards were not tested within the same experimental design as the others and so cannot be included within the ANOVA. However, for 11 of the 12 subjects, the qwerty speeds exceed that of all the other keyboards.)

Learning rate. It is possible that the learning of the alphabetical keyboard could proceed more quickly than that of the random or qwerty keyboard, because the structure of the keyboard would help in the learning of key placement. Two factors indicate that this is not likely to be the case. First, in using the standard touch-typing mapping of fingers to keys, one must learn which finger types which keys. This learning is not helped much by horizontally organized alphabetic keyboards. (For example, on alphabetical-1, the middle finger of the left hand types *c*, *m*, and *v*; the alphabetic nature of the keyboard arrangement is not apparent.) Second, the studies of Hirsch (1970) and Michaels (1971) showed no superiority in learning rates with the alphabetic keyboard.

We performed a simple analysis of differences in learning by comparing the number of keystrokes typed by each subject in the first half of each session with the number typed in the second half. In general, subjects did improve in the second half. However, the improvements were small; the differences in keystrokes between the second half and the first half were 18, 19, and 25 for the horizontal, diagonal, and random keyboards, respectively. Moreover, the standard deviations were relatively large: 15, 21, and 18, respectively. Thus, our results are consistent with those of the earlier studies. There appear to be no differences between the learning rates with the two alphabetic arrangements. Learning with the random keyboard is greater than that with the alphabetic, contrary to expectations. (It is also the case that typing rate on the random keyboard was slower; the greater learning rate could reflect the fact that the subjects have more to learn.)

Subjective preferences. Subjective preferences for keyboards varied. Three subjects said that they preferred the random board. Six said that they preferred the alphabetic boards, and three said they did not find much difference among the three. When asked to compare the two forms of alphabetic keyboards, alphabetical-1 (horizontal) and diagonal, two subjects said they had no preferences, five preferred the horizontal, and two preferred the diagonal. Two subjects said that the alphabetic keyboards were frustrating, whereas two subjects said that the random board took more concentration. One subject commented that although he performed better on the alphabetic keyboards, he would have preferred the random if he had had a chance to get used to it. In general, the subjective preferences do not appear to be useful. We did not expect this result; we believed that the subjects would prefer the alphabetically organized keyboards, even if their performance did not warrant it. Unfortunately, we took the preference rating at the end, after subjects had used all keyboards. Perhaps after 10 min of frustration in attempting to use the keyboards, their natural preferences had changed.

To test the influence of experience on subjective impressions, we selected new subjects in the same manner as we had selected the first 12. This group was asked to give their

subjective impressions of three keyboards (qwerty and the two alphabetical arrangements). In addition, their typing speeds on the qwerty keyboard were measured. The subjects were shown pictures of the three keyboards without being allowed any experience with them. There was still no difference in preference.

Conclusion. The appropriate assessment of alphabetic keyboards is made by comparing performance on them with performance on the random keyboard, thus avoiding contamination with the qwerty arrangement. Any negative influence of prior knowledge of the qwerty keyboard should have equal influence on both the alphabetic and the random keyboards. There is a slight advantage of the alphabetically organized keyboards over a randomly organized one. This advantage is around 10%, a difference of slightly less than one word per minute. Although the advantage is statistically significant, it is of no practical significance.

It is unfortunate that our subjects had some knowledge of the standard keyboard, even though we attempted to recruit subjects with no typing experience. It is difficult to find people who have had no experience with typewriters, expecially in a college environment. Nonetheless, the lack of meaningful difference between the alphabetic and the random keyboards is striking. Note the superiority of the qwerty keyboard for these subjects. Even though they could only type about one letter per second, the speed achieved on the qwerty keyboard was 67% greater than the speed reached with the alphabetic keyboards and 83% greater than that reached with the random keyboard.

THE EFFECT OF KEYBOARD ARRANGEMENT FOR EXPERTS

What would happen with expert typists? How would they be affected by different keyboard arrangements? We despaired of doing the actual experiments because to have done so would have required months of training on a variety of keyboards to get people to expert status, and then a considerable amount of retraining of the subjects so that they were back to normal on regular typewriters. The experiment has been done with the Dvorak keyboard. Performance estimates vary, ranging from a 5 to 10% improvement in typing speed to no significant difference (Alden et al., 1972).

It is possible to compute the effects of keyboard arrangement on typing speed. Kinkead (1975) devised a computational method based on the keystroke intervals for typing keystroke patterns, the relative frequency of English digraphs, and the physical arrangement of the keys. He estimated only a 2.6% improvement in typing speed with the Dvorak layout, although he thought it possible to devise other keyboard arrangements that could give a 7.6% improvement. That size improvement, he concluded, "would probably be lost in the flurry of retraining and rebuilding 20 million typists and machines" (Kinkead, 1975, p. 160).

Card, Moran, and Newell (in press) have extended Kinkead's analyses to cover other arrangements of keys. (They have also generalized and simplified the analytical methods.) They find that the Sholes keyboard fits in the middle of a range of possible speeds, with the Dvorak arrangement yielding a maximum improvement of 11% and certain inelegant alphabetic arrangements yielding a decrement of about 10% (Card and Moran, 1980).

Rumelhart and Norman (1982) have developed a computer simulation of the hand and finger movements of a skilled typist. With this program, it is possible to estimate the effects of any keyboard layout, including ones in which the keys are physically quite different from those of the standard typewriter. (The techniques of Kinkead [1975] and of

Card, et al. [in press] are tied to the standard typewriter by the use of empirical time estimates based on the qwerty layout.) This simulation includes the cognitive and motor control mechanisms, and the output is a computer-graphics display of the hands and fingers moving over the keyboard, plus the reaction time distributions of interkeystroke intervals. The model simulates the movements of the hands and fingers over the keyboard, and by using a parallel relaxation algorithm, allows for simultaneous movement of the fingers and hands toward the different letters of the word being typed, thus capturing the parallel, overlapping movements seen in high-speed films of expert typists. Detailed analyses and evaluation of the model and discussions of the overlapping motions of experts are available in Rumelhart and Norman (1982), Norman and Rumelhart (in press), and Gentner (in press).

Using this simulation model, we have tested a variety of keyboard layouts. The typing program was run with different keyboard layouts, using the same text as we used in our experiment. The typing model yields reaction time distributions for the interkeystroke intervals in arbitrary model units, and these were transformed into words per minute by assuming that each model unit was 40 ms. This yields a reasonable typing rate for an average, secretarial-level typist (about 55 words/min). In computing the typing speeds, carriage returns were not counted, but spaces and simple punctuation (periods and commas) were. The text was in all upper case, and so no shift-key operations were required. The data from the simulations are shown in Table 2. Despite the very different assumptions and mechanisms of the typing model from the other methods, our estimates agree with the work of Card and Moran (1980) within the accuracy of the estimations: about a 5% improvement for Dvorak, and a decrement for the horizontal alphabetic keyboards that ranges from 2% to 9%, dependent upon the particular layout that is used.

Different layouts of keyboards produce considerably different loadings of the two hands, as well as different percentages of keystrokes required off the home row. Thus, compare the results for alphabetical-1 and alphabetical-4 keyboards (Figure 1 and Table 2). The keyboards are very similar, with identical top rows, differing only in the placement of the letter *t* on the middle row, and having punctuation differences on the bottom row. Yet, this slight change in key placement leads to a considerable change in the amount of typing on the home row (38% versus 30%, in favor of alphabetical-4) and in the balance between the hands (with alphabetical-4 being more balanced). The simulation shows that alphabetical-4 is equal to the Sholes key-

TABLE 2

Typing Speeds Calculated from the Rumelhart and Norman (1982) Typing Model*

Keyboard	*% Row Use* *Top*	*Middle*	*Bottom*	*Space*	*% Left Hand*	*% Right Hand*	*Model words/min*	*% Deviation from qwerty*
Qwerty	42	27	15	16	57	43	56	—
Dvorak	21	56	7	16	47	53	58	+5.4
Alphabetical-1	37	30	17	16	64	36	52	−7.1
Alphabetical-2	29	29	24	16	54	46	55	−1.8
Alphabetical-3	43	18	23	16	66	34	51	−8.9
Alphabetical-4	37	38	9	16	57	43	55	−1.8
Diagonal	23	29	32	16	55	45	55	−1.8

* In these computations, the space key is assumed to be typed with the right thumb

board in speed, whereas alphabetical-1 is one of the slowest keyboards.

Detailed studies of the timing characteristics of typing have shown that between-hand, letter digraphs are typed faster than within-hand, and between-finger digraphs faster than within-finger (Gentner, in press; Kinkead, 1975; Rumelhart and Norman, 1982). These and other factors indicate that for optimum typing speed, keyboards should be designed so that (1) the loads on the right and left hands are equalized, (2) the load on the home (middle) row is maximized, (3) the frequency of alternating hand sequences is maximized, and (4) the frequency of same-finger typing is minimized.

The Dvorak keyboard does a good job on these criteria, especially the first two; 67% of the typing is done on the home row, and the left-right hand balance is 47% to 53%. Although the Sholes (qwerty) keyboard fails at Conditions 1 and 2 (most typing done on the top row and the balance between the two hands of 57% to 43%), the policy of putting successively typed keys as far apart as possible favors factors 3 and 4, thus leading to relatively rapid typing. Thus, the Sholes keyboard actually seems to be a sensible design, superior to all of the alphabetical arrangements that we have studied, and only 5 to 10% slower than the Dvorak keyboard, the one that was based on time-and-motion studies.

For the expert typist, the layout of keys makes surprisingly little difference. There seems no reason to choose Sholes, Dvorak, or alphabetically organized keyboards over one another on the basis of typing speed. It is possible to make a bad keyboard layout, however, and two of the arrangements that we studied can be ruled out. Note that one of the slower keyboards is "alphabetical-1." This configuration of keys matches well what people think of when considering alphabetical arrangements. It is not very good at all.

HOW COULD AN ALPHABETIC KEYBOARD HELP?

The experimental results for nontypists suggest that the alphabetic arrangement was not helpful. Why not? After watching ourselves and our colleagues type on the experimental keyboards, we agreed that an alphabetic arrangement seemed more rational: however, we did not seem to be able to use the alphabet in typing. One can verify this by attempting to type a line of text on each of the keyboards shown in Figure 1. (Type by pointing at the appropriate locations with the index finger.)

The alphabetically ordered keyboard helps only inasmuch as there is a correlation between letter position in the alphabet and key placement on the typewriter. Notice that the point at which the letters shift from one row to another is dictated by the physical arrangement of the keyboard and not by any logical structure of the alphabet. Suppose that one is typing the word *flower* on the alphabetical-1 keyboard. The *f* is located on the top row, near the middle of the keyboard; where should one look for the *l*? Knowledge of the alphabet indicates that the *l* is after the *f*, and so perhaps one should look to the right to find it. But whether or not this is true depends upon where the linear string of the alphabet is broken to fit it on the keyboard. In fact, on this keyboard one must look to the left to find the *l*. Knowledge of the alphabet does not correspond to knowledge of key location. Thus, a simple analysis of the alphabetic layout indicates that the alphabetical arrangement may not be so useful after all.

Now consider the "diagonal" arrangement shown in Figure 1. This arrangement of the letters allows position in the alphabet to guide placement of the keys. In the word *flower*, *l* occurs both later than *f* in the alphabet and to the right of it on the keyboard. The correlation between the position of the

letter in the alphabet and the location of the key is considerably higher for the diagonally organized alphabetic keyboard than for the horizontally organized keyboard. This keyboard arrangement should therefore be better than the horizontal arrangement. However, neither the experimental results nor our informal observations support this outcome.

How could the alphabet be of use, even for the diagonal keyboard? To be useful, a considerable amount of mental computation is required. Here is a basic algorithm for alphabet search.

(A1) Call the letter just typed the "current letter" and its keyboard location the "current location."
(A2) Determine the next letter to be typed from the text.
(A3) Mentally determine the position in the alphabet of the letter to be typed relative to that of the current letter.
(A4) Mentally translate the relative alphabetic position to a keyboard location; call the new keyboard location the "target location" (TL).
(A5) Do a saccadic eye movement from the current location to TL.
(A6) If the letter to be typed is at TL or in view, move the finger to the location and type the key, otherwise call the letter at TL the "current letter" and its keyboard location the "current location" and repeat from step A3.

Obviously, there is more to the use of an alphabetically organized keyboard than is obvious at first analysis. Compare how a person might type on a keyboard without making use of the alphabetic information (such as might be necessary with the "random" keyboard) with an algorithm based on visual search. Steps 1 and 2 are the same as for the alphabetic algorithm, but steps A3 through A6 are replaced with B3 and B4:

(B3) Make a saccadic eye movement from the current location to a new target location.
(B4) If the letter to be typed is at the target location (TL) or in view, move the finger to the location and type the key, otherwise call the letter at TL the "current letter" and its keyboard location the "current location" and repeat from step B3.

A major difference between the two algorithms is the amount of mental computation; Steps A3 and A4 for the alphabetical keyboard have no counterpart with the random keyboard. This means that there is a considerable amount of mental effort expended in the case of the alphabetic algorithm, almost none in the visual search algorithm. In addition, the alphabetic algorithm requires that typists be able to use information about the relative location of letters in the alphabet. Several studies, however, have shown that people do not know the relative locations of all pairs of the alphabet, but must compute them by a search of their long-term memory knowledge of the alphabet structure (see Hamilton and Sanford, 1978; Lovelace and Snodgrass, 1971). These analyses suggest why there might not be much aid given by alphabetically arranged keys; there is too much mental computation required and very little gain. In his recent review of studies of alphabetic keyboards Hirsch also reaches much the same conclusion:

> The alphabetical keyboard probably requires, first, a memory search to locate the letter in its approximate or relative alphabetical position, and then a visual search to find the key on the board (where it is situated without regard to its frequency of its use). Accordingly, the combination of the memory plus visual searches may be less efficient than a purely visual search where the probability is high that the visual area first scanned will contain the sought-for letter (Hirsch, 1981, p. 139).

CONCLUSION

In these studies we have analyzed the novice and expert use of different keyboards. First consider the implications of keyboard layout for novices. Our experimental studies of typing rates show that for beginners, alphabetically organized keyboards are slightly superior to a randomly organized one, but that this difference is too slight to be of any practical significance. Moreover, even a slight knowledge of the qwerty arrangement is apt to help considerably when using the Sholes

keyboard (and to interfere with the alphabetical one). We suspect that, on the whole, our subjects did *not* use the alphabetic algorithm, but instead resorted primarily to visual search on the keyboard, thus treating the key arrangement as if it were unstructured (random). As a comparison of the two different algorithms indicates, this reduces the mental workload considerably, with little effect on performance.

The Sholes keyboard may be easier to learn and to use than a random or alphabetic keyboard. Hirsch describes its virtues this way:

> Although not a perfect arrangement, the key arrangement of the standard typewriter is also not a random one. Whatever its limitations, it was "human engineered" even as early as 1873, and many of the most frequently used letters are, generally speaking, clustered in the center of the keyboard. Hence, hunting for a letter can usually be confined to a relatively small visual area (Hirsch, 1981, p. 139).

Now consider the implications for expert typists. The appropriateness of the simulation studies of different keyboard arrangements for expert typists depends upon the validity of the simulation program. In our detailed examination of the simulation model for the Sholes keyboard, it seemed to provide an accurate simulation of many of the properties of the reaction-time distributions of skilled typists (Rumelhart and Norman, 1982). In addition, the results presented here for the comparison of different keyboards agree reasonably well with other methods of computation and with experiments comparing Dvorak and Sholes keyboards. These results, therefore, seem reasonable, if only because they are consistent with other methods.

Finally, our analysis of the operations a novice typist would have to perform in order to take advantage of the alphabetical arrangement of keys provides a possible explanation for the lack of improvement with the introduction of these keyboards; the mental load required to make use of the keyboard is too great. (And it is not clear that most people know the alphabet well enough to take full advantage of the ordering.)

The conclusion is that both experimental results and various forms of computational models agree that it is a waste of time to rearrange the typewriter layout; there are too few savings to be had. However, this does not mean that typewriter layout cannot be improved. Kinkead (1975) pointed out that considerable savings could be made by either moving the return key or by eliminating it (with computer-assisted text editing). Elimination of the return key gives a minimum of 7% improvement in speed, and "up to 30% when the original copy is not properly formatted" (Kinkead, 1975, p. 160). Thus, simple changes in the way that typists use the keyboards can have a large effect without changing the key layout.

There is considerable room for dramatic revision of typewriter structure (see Litterick, 1981, for a quick review). There is little justification for the present size of the keyboard, for the staggered arrangement of its keys, and for other aspects of its physical arrangements. The left and right hands should have a mirror-image configuration of keys, rather than the continual upper-left to bottom-right diagonal slope of the keys in the present arrangement. The left- and right-hand portions of the keyboard could be separated, allowing the manuscript that is being copied to be placed between the hands, where the typist can see it without twisting his or her head. The space bar seems wasteful; most skilled typists only use the right thumb to type spaces, and then only use a small section of the bar. Alternatively, the keyboards could be sensitive to both upward and downward motion; keyboards could allow (require) typing of several overlapping keys, or simultaneous depressing of several keys (these are called "chord" keyboards); several keystrokes could

be required per letter, thereby allowing one to reduce the numbers of keys. Using this technique Gopher and Eilam (1979) have developed a successful one-hand keyboard for Hebrew. Many of these ideas have been tested in various formal and informal experiments. We have not seen a proper assessment of them, however, comparing typing speeds, error rates, and difficulty of learning. The standard "chord keyboard" has been studied, and although it is indeed very fast (which is why it is used by court recorders), it is not easy for a novice. A review of many of these methods is found in Alden et al., 1972. Our lesson is simply this: Do not waste time rearranging the letter arrangement of the existing standardized keyboard.

ACKNOWLEDGMENTS

This research was performed under Contract N00014-79-C-0323, NR 157-437 with the Personnel and Training Research Programs of the Office of Naval Research, and was sponsored by the Office of Naval Research and the Air Force Office of Scientific Research. We gratefully acknowledge the advice of Stu Card and Tom Moran of the Xerox Palo Alto Research Center, and the technical assistance of Donald Gentner and Phil Mercurio.

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